Mathematical Programming — Project Report

Portfolio Optimization via Stochastic Linear Programming and Conditional Value-at-Risk

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Abstract

We explore the evolution of portfolio optimization from Markowitz's original quadratic programming formulation (1952) to modern stochastic linear programming approaches based on Conditional Value-at-Risk (CVaR). We trace the historical development through linearization attempts (MAD model, 1991), the introduction of downside risk measures (VaR), and the breakthrough result of Rockafellar and Uryasev (2000): CVaR can be minimized via linear programming. The key insight is that Sample Average Approximation transforms the stochastic CVaR optimization into a deterministic linear program. We provide the mathematical formulation and discuss extensions to mixed-integer programming for cardinality-constrained portfolios.

Portfolio Optimization: Problem Formulation

An investor must allocate a fixed capital W across n assets (e.g., stocks, options, futures). Let x_i denote the fraction of capital invested in asset i, with $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.

Future asset returns are uncertain, represented by random variables $\xi = (\xi_1, \ldots, \xi_n)$. The portfolio return is:

$$R = \sum_{i=1}^{n} x_i \xi_i.$$

The investor faces two competing objectives:

- Maximize expected return, namely $\mathbb{E}[R]$;
- Minimize risk, for which we need to introduce some measure of uncertainty in R.

For the model to be meaningful, we impose the following constraints:

$$\sum_{i=1}^{n} x_i = 1 \quad \text{(budget)}$$

$$x_i \geq 0 \quad \text{(no short-selling, optional)}$$

Portfolio optimization seeks to find efficient portfolios that offer the best possible return for a given level of risk, or equivalently, the lowest risk for a given target return.

The mathematical difficulty lies in:

1. Modeling uncertainty in future returns;
2. Quantifying risk in a mathematically tractable way;
3. Solving the resulting optimization problem efficiently.

Different choices for the risk measure and treatment of uncertainty lead to different optimization formulations and computational approaches.

From Quadratic Programming to Stochastic Linear Programming: the evolution of Portfolio Optimization

The original problem and its limitations

In 1952, Harry Markowitz formulated the portfolio optimization problem as a quadratic programming problem. Given n assets with expected returns $\mu_i = \mathbb{E}[\xi_i]$ and covariances $\Sigma_{ij} = \mathrm{Cov}(\xi_i, \xi_j)$, the investor chooses portfolio weights $x = (x_1, \ldots, x_n)$ to minimize variance for a target expected return:

$$
\begin{aligned}
\text{minimize} \quad & x^T \Sigma x \\
\text{subject to} \quad & \mu^T x \geq \rho, \\
& \sum_{i=1}^{n} x_i = 1, \quad x \geq 0, \\
& 0 \leq x_i \leq u_i, \qquad i = 1, \ldots, n,
\end{aligned}
$$

where $\mu = (\mu_1, \ldots, \mu_n)$, ρ is a parameter representing the minimal rate of return required by an investor and u_i is the maximum amount of money which can be invested into asset i.

This approach presents two fundamental criticalities. The first is computational: in first place, we have to calculate $n(n+2)/2$ constants $\Sigma_{i,j}$. Furthermore, solving quadratic problems with constraints becomes rapidly intractable as the number of assets increases. The second is conceptual: variance is a *symmetric* risk measure: it penalizes positive deviations (unexpected profits) and negative ones (losses) equally, when in reality investors are primarily concerned with losses.

In summary, large-scale portfolio optimization using Markowitz's (full covariance matrix) model has been considered impractical not only because of its computational difficulty but also because of the complications associated with the implementation of its solution.

From quadratic to linear: the MAD model

The computational burden of quadratic programming motivated the search for linear alternatives. In 1991, Konno and Yamazaki proposed the MAD model (Mean Absolute Deviation), replacing variance with a different measure for the risk:

$$
w(x) = \mathbb{E}\left[\left|\sum_{j=1}^{n} \xi_j x_j - \mathbb{E}\left[\sum_{j=1}^{n} \xi_j x_j\right]\right|\right]
$$

The optimization problem becomes:

$$
\begin{aligned}
\text{minimize} \quad & w(x) \\
\text{subject to} \quad & \mu^T x \geq \rho, \\
& \sum_{i=1}^{n} x_i = 1, \quad x \geq 0, \\
& 0 \leq x_i \leq u_i, \quad i = 1, \ldots, n.
\end{aligned}
\tag{1}
$$

Through linearization techniques this transforms into a linear program: let r_{jt} be the

realization of random variable ξ_j during period t ($t = 1, \ldots, T$) which we assume to be available through the historical data or from some future projection. We also assume that the expected value of the random variable can be approximated by the average derived from these data.

In particular, let

$$r_j := \sum_{t=1}^{T} r_{jt}/T \approx \mathbb{E}[\xi_j].$$

Then $w(x)$ can be approximated as follows:

$$\mathbb{E}\left[\left|\sum_{j=1}^{n} \xi_j x_j - \mathbb{E}\left[\sum_{j=1}^{n} \xi_j x_j\right]\right|\right] \approx \frac{1}{T}\sum_{t=1}^{T}\left|\sum_{j=1}^{n}(r_{jt} - r_j)x_j\right|.$$

Let us denote

$$a_{jt} = r_{jt} - r_j, \quad j = 1, \ldots, n; \quad t = 1, \ldots, T.$$

Then (1) leads to the following minimization problem:

$$\begin{aligned}
\text{minimize} \quad & \sum_{t=1}^{T}\left|\sum_{j=1}^{n} a_{jt}x_j\right|/T \\
\text{subject to} \quad & \sum_{j=1}^{n} r_j x_j \geq \rho, \\
& \sum_{j=1}^{n} x_j = 1, \\
& 0 \leq x_j \leq u_j, \quad j = 1, \ldots, n,
\end{aligned}$$

which is equivalent to the linear program

$$\begin{aligned}
\text{minimize} \quad & \sum_{t=1}^{T} y_t/T \\
\text{subject to} \quad & y_t + \sum_{j=1}^{n} a_{jt}x_j \geq 0, \quad t = 1, \ldots, T, \\
& y_t - \sum_{j=1}^{n} a_{jt}x_j \geq 0, \quad t = 1, \ldots, T, \\
& \sum_{j=1}^{n} r_j x_j \geq \rho, \\
& \sum_{j=1}^{n} x_j = 1, \\
& 0 \leq x_j \leq u_j, \quad j = 1, \ldots, n.
\end{aligned}$$

However, MAD still penalizes deviations above and below the mean equally. The conceptual limitation of Markowitz's framework persists.

Value-at-Risk: the right measure with the wrong tools

In the 1990s, the financial industry adopted Value-at-Risk (VaR) to measure risk associated with portfolio selection. VaR at confidence level α is defined as:

$$\mathrm{VaR}_\alpha(x) = \inf_\xi \left\{ v : \mathbb{P}(f(x,\xi) > v) \leq 1 - \alpha \right\}$$

where f is a function representing the loss associated with the decision variables x and the uncertain assets return ξ. For example, $\mathrm{VaR}_{0.95}(x) = 0.10$ means "choosing portfolio x, with 95% probability, the loss will not exceed 10%." It is exactly what is needed: an asymmetric measure (looks only at losses), quantile-based (captures extreme events), and naturally stochastic (defined with respect to a probability distribution).

The problem? VaR is not convex as a function of the portfolio weights x. This makes optimization numerically unstable and without guarantees of convergence to global optima.

The breakthrough of Rockafellar and Uryasev (1999-2000)

Rockafellar and Uryasev introduced the Conditional Value-at-Risk (CVaR) ([RU00]) defined as:

$$\mathrm{CVaR}_\alpha(x) = \mathbb{E}[f(x,\xi) \mid f(x,\xi) \geq \mathrm{VaR}_\alpha(x)].$$

Intuitively, if we order all possible loss scenarios from worst to best, VaR is the threshold of the worst $(1-\alpha)$ fraction, while CVaR is the average loss among those worst-case scenarios. For a fixed α, our model now becomes:

$$\begin{aligned} \text{minimize} \quad & \mathrm{CVaR}_\alpha(x) \\ \text{subject to} \quad & \mu^T x \geq \rho, \\ & \sum_{i=1}^{n} x_i = 1, \quad x \geq 0, \\ & 0 \leq x_i \leq u_i, \quad i = 1, \ldots, n. \end{aligned} \tag{2}$$

Their fundamental result is twofold:

1. CVaR is convex as a function of the portfolio x, when f is convex with respect to x, even though VaR is not

2. CVaR can be optimized via a linear program through the representation:

$$\mathrm{CVaR}_\alpha(x) = \min_\zeta \left\{ \zeta + \frac{1}{1-\alpha}\mathbb{E}[(f(x,\xi) - \zeta)^+] \right\},$$

where $(z)^+ = \max(z, 0)$.

This leads to the following formulation equivalent to (2)

$$
\begin{aligned}
\min_{\zeta,x} \quad & \left\{\zeta + \frac{1}{1-\alpha}\mathbb{E}[(f(x,\xi)-\zeta)^+]\right\} \\
\text{subject to} \quad & \mu^T x \geq \rho, \\
& \sum_{i=1}^{n} x_i = 1, \quad x \geq 0, \\
& 0 \leq x_i \leq u_i, \quad i = 1,\ldots,n.
\end{aligned} \tag{3}
$$

Assuming f to be convex, the problem of minimizing CVaR is therefore one of convex programming.

From stochastic to deterministic problem: Sample Average Approximation

The original CVaR model is stochastic because it contains an expectation $\mathbb{E}[(f(x,\xi)-\zeta)^+]$ over continuous random variables.

Sample Average Approximation (SAA) resolves this by approximating the expectation with the sample mean over N scenarios $\{\xi^1,\ldots,\xi^N\}$ generated via Monte Carlo sampling:

$$
\zeta + \frac{1}{1-\alpha}\mathbb{E}[(f(x,\xi)-\zeta)^+] \approx \zeta + \frac{1}{N(1-\alpha)}\sum_{k=1}^{N}(f(x,\xi^k)-\zeta)^+
$$

For portfolio optimization with losses $f(x,\xi^k) = -(\xi^k)^T x$, introducing auxiliary variables u_k, the stochastic problem becomes a deterministic linear program:

$$
\begin{aligned}
\text{minimize} \quad & \zeta + \frac{1}{N(1-\alpha)}\sum_{k=1}^{N} u_k \\
\text{subject to} \quad & u_k \geq -(\xi^k)^T x - \zeta, \quad k = 1,\ldots,N \\
& u_k \geq 0, \quad k = 1,\ldots,N \\
& \sum_{i=1}^{n} x_i = 1, \quad x \geq 0, \\
& \mu^T x \geq \rho, \\
& 0 \leq x_i \leq u_i, \quad i = 1,\ldots,n.
\end{aligned}
$$

This is a standard LP with $n + N + 1$ variables and $2N$ inequality constraints.

The transformation from stochastic problem to deterministic LP via SAA is the key computational breakthrough that makes CVaR optimization tractable using standard linear programming methods.

Adding integer constraints: Stochastic Integer Programming

In practice, real portfolios require discrete constraints:

- Cardinality constraints: invest in at most K assets (to reduce complexity and costs)
- Lot sizes: minimum/discrete investment quantities
- Transaction costs: fixed costs for buying/selling

These constraints are modeled with binary variables $y_i \in \{0, 1\}$ where $y_i = 1$ if asset i is selected, $y_i = 0$ otherwise:

$$
\begin{aligned}
\text{minimize} \quad & \zeta + \frac{1}{N(1-\alpha)} \sum_{k=1}^{N} u_k \\
\text{subject to} \quad & u_k \geq -(\xi^k)^T x - \zeta, \quad k = 1, \ldots, N \\
& u_k \geq 0 \\
& \sum_{i=1}^{n} y_i \leq K \quad \text{(at most } K \text{ assets)} \\
& \sum_{i=1}^{n} x_i = 1, \quad x \geq 0 \\
& x_i \leq y_i \quad \text{(linking constraints)} \\
& 0 \leq x_i \leq u_i, \quad i = 1, \ldots, n, \\
& y_i \in \{0, 1\}
\end{aligned}
$$

The linking constraints $x_i \leq y_i$ enforce that $x_i = 0$ when $y_i = 0$ (asset not selected) and allow $x_i > 0$ only when $y_i = 1$ (asset selected). The problem becomes a Mixed-Integer Linear Program (MILP).

The final synthesis

The final result is powerful: problems that were nonconvex stochastic optimization (intractable) become standard MILPs solvable with commercial solvers, while maintaining theoretical correctness and practical relevance (integer constraints, thousands of scenarios). The paper by Rockafellar and Uryasev is the turning point that makes all of this possible: it provides both the right risk measure (CVaR) and the right mathematical framework (convex optimization via LP) that enables the integration of stochastic programming and integer constraints in a computationally tractable way.

References

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